

s needlessly

(e.g. <https://gitlab.com/gitlab-org/gitlab/-/issues/285597>). Large files (LFS, packages, etc.) are slow to finalize or don't work at all as a result.

Definitions

CarrierWave

A gem that provides a simple and extremely flexible way to upload files from Ruby applications. This was the boring solution when first implemented. However this is no longer our use-case, as we upload files from Workhorse, and we had to patch CarrierWave's internals to support Direct Upload.

Direct upload

A technology we developed to intercept file uploads with Workhorse and handle the expensive upload operation in Workhorse, where it's cheaper. See our uploads development documentation for more details.

Kickoff video

{{< youtube "X9Vw8hsM8E" >}}

Exit criteria (100%)

The overarching goal should be to define improvements that can be made with the Object Storage implementation, and make informed implementation proposals through the work of this group. As such we intend to:

- Document the status quo of Object Storage and classify its use by feature vertical and integration patterns,
 - since we know there to be drift between features.
- Categorize existing object storage buckets
- Describe the current state of Object Storage implementation
- Outline a path forward by designing a new simplified architecture for Object Storage. Identify high-level steps we need to take for that architecture to be realized.
 - Requirements for a the new Object Storage architecture
 - Proposal: unified blob storage
 - Object Storage: storing attachments without carrierwave
- Prototype individual aspects of the proposed architecture by exploring both new technology such as ActiveStorage, or by reworking existing code.
 - Document and refactor Workhorse upload routines
 - POC: single authorization endpoint
 - POC: ActiveStorage experiment
- Identify features and configuration that should be deprecated to reduce the maintenance complexity.

- Deprecate background upload
- Object Storage: remove background upload
- Evaluate deprecation and removal of the Pseudonymizer
- Deprecate and remove Pseudonymizer

Out of scope

- Make final decisions on proposed solutions.
- Implement all proposed solutions.
- Be a permanent custodian for or oversee Object Storage development in the future.

Outcome

At the beginning of this working group, we had three main areas of improvement: consolidating object storage files into a single bucket, reducing code complexity, and removing local storage.

However, it took us very little time to figure out that the biggest challenge for the working group members was understanding the current implementation and being able to speak a common language.

The working group led an effort to collect and categorize all the usages of object storage in the product with the result of building a shared understanding of the problem, producing a renewed Uploads Development Guide, and removing features such as Pseudonymizer and background uploads.

Consolidating object storage files into a single bucket and removing local storage support were assessed by the working group as excellent ways to reduce code complexity and simplify the product installation and maintenance. However, those topics require more significant cross-department decisions that do not fit the working group's scope. As a first iteration, the working group members addressed how to reduce code complexity by focussing on technological challenges.

The creation of the scalability frameworks team during this working group execution provided a perfect partner to give continuity to this effort. [Epic gitlab-com/gitlab-infra&733](https://gitlab-com/gitlab-infra/-/issues/733) describes the current roadmap.

Roles and responsibilities

The functional leads will be responsible for:

- Representing the needs of individual stakeholders in their department/sub-dept.
- Gathering and consolidating feedback on specific proposals from their department/sub-dept.
- Communicating the output from the working group (if any) and answering questions from their dept/sub-dept.

Ideally, the functional lead is someone who is an IC working in the affected groups, but anyone

capable of representing a group, department, or sub-department in the fashion mentioned above is welcome.

Working Group	Role	Person	Stakeholder Dept.	Title
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Executive Sponsor		Marin Jankovski @marin	Infrastructure	Director of Infrastructure, Platform
Facilitator		Alessio Caiazza @nolith	Infrastructure	Staff Backend Engineer
Functional Lead		Grzegorz Bizon @grzesiek	Ops, Verify	Staff Backend Engineer
Functional Lead		Jason Plum @WarheadsSE	Distribution	Staff Backend Engineer
Functional Lead		Matthias K��ppler @mkaeppler	Memory	Senior Backend Engineer
Functional Lead		ukasz Korbasiewicz @lkorbasiewicz	Support	Support Engineer
Member		Vladimir Shushlin @vshushlin	Release group	Senior Backend Engineer
Member		Erick Bajao @iamricecake	Verify	Senior Backend Engineer
Member		Jaime Martinez @jaime	Package	Backend Engineer
Member		David Fernandez @10io	Package	Senior Backend Engineer
Member		Tiger Watson @tigerwnz	Configure	Senior Backend Engineer
Member		Vitor Meireles De Sousa @vdesousa	AppSec	Senior Application Security Engineer
Member		Patrick Bajao @patrickbajao	Workhorse	Senior Backend Engineer
Member		Catalin Irimie @cat	Geo	Senior Backend Engineer
Member		Sofia Vistas @svistas	Quality	Senior Software Engineer in Test
Member		Jacob Vosmaer @jacobvosmaer-gitlab	Scalability	Staff Backend Engineer

Company efforts on uploads

At GitLab we work in iterations, direct upload was developed by several teams incrementally by adding new features over the course of several milestones.

To demonstrate the number of teams and milestones involved, the timeline of the Object Storage development, from feature development to tech debt and security fixes, is outlined:

- From 10.4 to 10.5 the CI/CD team worked on a product discovery to figure out how we could remove the NFS storage to allow the release of a cloud native GitLab.
- In 10.6 the CI/CD team developed direct upload for Git LFS.
- In 10.7 the plan team ported Object Storage support from EE to CE.
- In 10.7 the CI/CD team implemented direct upload for artifacts, only on Google Cloud Storage.
- In 10.8 and 11.0 the CI/CD team implemented direct upload for artifacts on AWS S3 and compatible storages.
- In 11.1 the plan team implemented direct upload for user attachments.
- In 12.5 the delivery team worked on some technical debt, this was a blocker for our 50k reference architecture as well as causing several problems on GitLab.com infrastructure, like VMs reboots, LFS Access Forbidden, and problems with the new pre environment.
- In 12.6 we fixed a P1/S1 security issue - Path traversal leading to Remote Code Execution. After the fix, P2/S2 variations of the original issues where reported:
 - In 12.7 and 12.8 we fixed a path traversal in dependency proxy.
 - In 12.10 we fixed a path traversal in NPM packages.
 - In 13.0 we fixed a path traversal in Nuget packages.
 - In 13.5 we fixed a P1/S1 path traversal on Git LFS.
- In 12.7 we fixed a P1/S1 security issue - Workhorse bypass leads to package disclosure and file disclosure in /tmp. After the fix, P1/S1 variations of the same attack where reported:
 - In 12.10 we fixed two workhorse bypasses, one for Nuget packages and one for artifacts.
 - Another security release, still in 12.10, was needed to completely fix Nuget packages.
 - In 13.2 we fixed a workhorse bypass in Maven repository.
 - In 13.3 we fixed a workhorse bypass in Conan repository.
- In 13.1 an IC added consolidated object storage configuration.
- In 13.2 an IC added a S3 client to Workhorse to improve multipart uploads and ETag checking.
- In 13.3 an IC added support for AWS S3 server side encryption.
- In 13.4 an IC added support for Azure Blob storage.
- From 13.4 to 14.0 the Release group transitioned GitLab Pages from NFS to Object Storage following the GitLab Pages New Architecture blueprint

SECTION: company/working-groups/performance-indicators.md

Attributes

Property	Value
Date Created	May 30, 2019
End Date	August 1, 2019
Slack	wgperf-indicators (only accessible from within the company)
Google Doc	Performance Indicator Working Group Agenda (only accessible from within the company)
Issue Board	Performance Indicator Working Group Board

Business Goal

Erect automated KPI data collection and visualization in order to manage-at-scale.

Exit Criteria

- Get "Engineering Pulse Survey" KPI to maturity level 3
- Get "Hiring actual vs plan" KPI to maturity level 4
- Get "Non-headcount budget vs plan" KPI to maturity level 4
- Get "Average location factor" KPI to maturity level 4
- Get "Diversity" KPI to maturity level 3
- Get "Handbook updates" KPI to maturity level 4
- Get "Employee retention" KPI to maturity level 3
- Move KPI overview from Google doc to the handbook page(s)

Roles and Responsibilities

Working Group Role	Person	Title
Facilitator	Eric Johnson	VP of Engineering
Data Lead	Walter Zabaglio	Director of Business Operations
PO Lead	Britany Rohde	Compensation & Benefits Manager
Finance Lead	Brooks Royer	Finance Business Partner
Member	Emilie Schario	Data Engineer, Analytics
Member	Taylor A. Murphy, PhD	Staff Data Engineer, Architecture
Member	Christopher Lefelhocz	Senior Director of Development
Member	Christie Lenneville	Director of UX
Executive Stakeholder	Paul Machle	CFO

SECTION: company/working-groups/pipeline-validation-service-operations.md

Attributes

Property	Value
Date Created	2021-06-03
Target End Date	2021-12-03
Slack	fpipelinevalidationservice (only accessible from within the company)
Google Doc	Agenda (only accessible from within the company)
Boards	gitlab-org board, gitlab-com board
Issue Label	~WorkingGroup::PipelineValidationServiceOperations

Goals

This Working Group has the following goals:

1. Operate and develop the Pipeline Validation Service (PVS)
1. Track PVS workload and effectiveness
1. Transition PVS to a long term owner

Exit Criteria

1. Successfully transition PVS to a long term owner

- The Anti-abuse group owns and maintains PVS

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	Christopher Lefelhocz	VP Development
Facilitator APAC	Mon Ray	Engineering Manager anti-abuse
Facilitator AMER	Wayne Haber	Director Engineering
Development Functional Lead	Stan Hu	Engineering Fellow
Security Functional Lead	Charl de Wit	Security Manager, Trust and Safety
Security Functional Lead	Roger Ostrander	Senior Security Engineer, Abuse Operations
Infrastructure Functional Lead	Igor Wiedler	Staff Site Reliability Engineer
Product Functional Lead	Taylor McSastlin	Product Manager, anti-abuse
Member	Sam Goldstein	Director Engineering, Ops
Member	Grzegorz Bizon	Staff Backend Engineer, Verify
Member	Joanna Shih	Quality Engineering Manager, Ops
Member	Alex Groleau	Interim Manager, Security Automation
Member	Jackie Porter	Group Manager, Product, Verify

SECTION: company/working-groups/product-accessibility.md

Attributes

Property	Value
Date Created	2023-07-19
Target End Date	TBD
Slack	wgproductaccessibility (only accessible from within the company)
Google Doc	Agenda (only accessible from within the company)
Epic	HQ epic
Overview & Status	See Exit Criteria Progress

Context

Since we were able to create a proof of concept with axe-core to automate accessibility checks in feature specs, we would like to include these assertions in the most of the GitLab feature specs, as well as in GitLab UI components codebase. This will allow us to ensure that GitLab accessibility compliance covers at least 57% of the WCAG guidelines (percent of accessibility issues that can be automated). It will also let us shift left our efforts to provide an accessible platform, by giving developers an easy way to monitor how introduced changes influence compliance. This initiative will entail writing tests, understanding existing

violations, and finding ways to fix or document them.

Stakeholders for this project include frontend engineers, backend engineers and product designers.

Exit Criteria

This Working Group has the following goals:

1. Include axe automated accessibility checks in automated feature tests.
1. Include axe automated accessibility checks in GitLab UI and write tests for components.
1. Compile a comprehensive list of known accessibility violations.
1. Develop and implement a plan for addressing accessibility violations, including evaluation plan/process and tooling.
1. Identify and bridge the gaps in accessibility frontend guide.
1. Plan for internal training.

Exit Criteria Progress

Criteria				
Start Date	Completed Date	Progress	DRI	

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Add axe-core checks to feature specs				15-08-2023
20%	@psjakubowska			
Add axe-core checks to components specs				16-04-2024
23-05-2025	100%	@rchanila		
Compile a comprehensive list of known accessibility violations				19-10-2024
03-01-2025	100%	@tauriedavis		
Develop and implement a plan for addressing accessibility violations, including evaluation plan/process and tooling	04-10-2024	30%	@psjakubowska	
Identify and bridge the gaps in accessibility frontend guide.				12-10-2023
27-02-2025	100%	@psjakubowska		
Plan for internal training	29-02-2024	100%	@jeldergl	

Roles and Responsibilities

Working Group Role	Person	Title
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Executive Sponsor	Tim Zallmann	VP of Engineering, Core Development
Facilitator		
Functional Lead	Paulina S-d-ak-Jakubowska	Frontend Engineer, Monitor:Respond
Functional Lead	Rahul Chanila	Senior Frontend Engineer, Package
Functional Lead	Jeremy Elder	Staff Product Designer, Manage:Foundations

Member	Lisa Falcone	Senior Paralegal, Legal & Corporate Affairs
Member	Scott de Jonge	Senior Frontend Engineer,
Manage:Foundations		
Member	Rajan Mistry	Frontend Engineer, Plan:Product Planning
Member	Deepika Guliani	Senior Frontend Engineer, Plan:Project
Management		
Member	Mohamed Moustafa	Fullstack Engineer, Fulfillment:Provision

SECTION: company/working-groups/product-analytics-gtm.md

Attributes

Property	Value	
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Date Created	March 10, 2020	
End Date	2022-03-10	
Slack	wg-gtm-product-analytics (only accessible from within the company)	
Google Doc	Working Group Agenda (only accessible from within the company)	
Quarterly Updates	FY21-Q4 FY22-Q1 FY22-Q2 EOQ Review	
Related Overview	GTM Product Analytics: Status, Gaps and the Road Forward	
Associated OKRs	FY21-Q2 - Architect Telemetry, FY21-Q3 - Deploy Product Analytics, FY21-Q4 - Deploy Product Analytics in GTM Motion	

Problem To Solve

Integrate GitLab product usage data and related data into CRO organization business processes to accelerate customer adoption and improve customer retention.

Business Goal

Embed product usage and related data across GTM business processes and systems (Salesforce, Gainsight, Marketo, SiSense, etc.).

Exit Criteria

KR: Deploy license usage in Salesforce and Gainsight for self-managed customers with usage ping data received and 99% of SaaS

KR: Automate basic metrics for Create, Verify, and Secure within Salesforce and Gainsight to track and drive adoption and expansion

KR: Ensure delivery of basic Salesforce and Gainsight enablement to GTM teams

For a more extensive resource, see GTM Product Analytics: Status, Gaps and the Road Forward.

Step 1: Understand what is needed and Architect Product Analytics = 100%

Epic

- · Collect Sales Specific feedback on Usage Ping)
- · Design and send Master Subscription schema table to Salesforce
- · Develop basic architectural plans for mapping data to Salesforce

Step 2: Deploy Telemetry => 90%

Epic

- · Deploy License Usage to Salesforce
- · Identify the top metrics for Create, Verify, Secure to drive account adoption and expansion
- · Create Metrics Dictionary for Usage Ping - Metrics Dictionary
- · Link usage data with subscription and account data
- · Determine Wave 3 Product Analytics metrics
- · Identify top data quality blockers - issues have been identified for Product Analytics and Fulfillment
- · Deploy license usage in Salesforce and Gainsight for self-managed customers with usage ping data received and 99% of SaaS

Step 3: Deploy Product Analytics across entire GTM Motion => 40%

Epic

- · Deploy from Sisense to Salesforce and Gainsight (internal link)
- · Automate basic metrics for Create, Verify, and Secure within Salesforce and Gainsight to track and drive adoption and expansion
- · Enable GTM teams with best practices

Roles and Responsibilities

Working Group Role	Person	Title
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Executive Stakeholder	David Sakamoto	VP Customer Success
Facilitator Emily McInerney	Manager Customer Success Operations	
Functional Lead, Customer Success Operations		Jeff Beaumont
Director Customer Success Operations		
Functional Lead, Data	Rob Parker	Senior Director, Data and Analytics
Functional Lead, Product Analytics	Amanda Rueda	Interim PM, Product Intelligence
Functional Lead, Fulfillment	Justin Farris	Group Manager, Product - Fulfillment
Functional Lead, Sales Systems	Jack Brennan	Director Sales Systems
Functional Lead, Product Intelligence	Nicolas Dular	Engineering Manager, Product Intelligence
Functional Lead, Product Usage	Sushma Nalamaru	Senior Data Engineer

Member	Jim Petr	Business Systems Engineer
Member	Jake Bielecki	Senior Director, Sales
Strategy & Analytics		
Member	Caitlin Ankney	Customer Success
Operations Analyst		
Member	Melia Vilain	Manager, Sales Analytics
Member	Brandon Butterfield	Customer Success
Operations Analytics Analyst		
Member	Justin Stark	Staff Data Engineer
Member	Michael Arntz	Customer Success Strategy
Manager		

Meetings

Meeting notes are logged in the Weekly Meeting Notes.

Working Group Closure

1. We've achieved our original exit criteria
1. We've built a reliable working model between Customer Success, Product & Engineering, and the Data team. We no longer need a formal working group to maintain progress
1. Work on this initiative will continue via the top 12 initiatives

SECTION: company/working-groups/product-analytics.md

Attributes

Property	Value
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Date Created	May 17, 2019
Date Ended	TBD
Slack	wgtelemetry (only accessible from within the company)
Google Doc	Product Analytics Working Group Agenda (only accessible from within the company)

Business Goal

Fast track GitLab to becoming a data-driven organisation by meeting the data needs of the Product team. Specifically:

- Enable Product to measure stage-specific usage activity, including read activity (frontend; a user in GitLab views a page) and write activity (backend; an object is created)
- Track stage-specific health metrics like SMAU.

Exit Criteria

- Create a Periscope dashboard for a single stage that measures SMAU and feature use on

GitLab.com.

- Ensure we can replicate this for other stages with great documentation.

Roles and Responsibilities

Working Group Role	Person	Title
Functional Lead	Luca Williams	Product Manager, Fulfillment
Functional Lead	Bartek Marnane	Dir of Engineering, Growth
Functional Lead	Dave Smith	Manager, Reliability Engineering
Executive Stakeholder	Scott Williamson	VP of Product
Member	Donald Cook	FE Engineering Manager, Plan
Member	Jeremy Watson	Sr Product Manager, Manage
Member	Walter Zabaglio	Dir of Business Operations
Member	Eli Kastelein	Data Analyst, Growth

SECTION: company/working-groups/product-career-development-framework.md

Attributes

Property	Value
Date Created	April 21, 2021
Target End Date	January 31, 2023
Slack	wgproduct-cdf (only accessible by GitLab team members)
Google Doc	Working Group Agenda (only accessible by GitLab team members)
Docs	Framework Handbook Page
Issues	Issue List / Issue Board
Label	~wg-product-cdf
Associated KPIs/OKRs	Promotion Rate and Retention
GitLab group for working group	@wg-product-cdf

Business Goals

1. Deliver a revised Product Career Development Framework for Individual Contributor roles in Product based on new competencies by 2022-06-20.

1. Conduct AMAs and training with Product Organization on new CDFs for optional use in a FY23-Q3 Career Conversations.

1. Conduct AMAs and training with Product Organization on new CDFs for broad use in FY23-Q4 Career Conversations.

Protocols and Processes

- 1. We will have a recurring weekly sync with the working group
- 1. We will use a issue board for assigning tasks and delegating efforts across the working group
- 1. We will track efforts in phases, which will be broken up into epics
- 1. The DRI for the CDF is the VP of Product Management and all changes for the CDF will be approved via the VP

Exit Criteria

- 1. Research and incorporation of industry standards into Product Manager Skills and Requirements => 100% delivered in Product2460
- 1. Research and incorporation of internal expectations into Product Management Responsibilities => 100% delivered in Product2517 and Product2832
- 1. Clear quantative measures for progress included in the CDF
- 1. Training for how to use the CDF for all Product Leaders
- 1. Simplification of PM Competencies

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	David DeSanto	VP, Product
Facilitator	Jackie Porter	Director of Product, Verify & Package
Facilitator	Farnoosh Seifoddini	Group Product Manager, Product Operations
Functional Lead	Kenny Johnston	Senior Director of Product, Ops & Analytics
Functional Lead	Joshua Lambert	Director of Product, Enablement
Functional Lead	Hillary Benson	Director of Product, Sec & Data Science
Functional Lead	Justin Farris	Sr Director, Product Monetization
Reviewer	Gabe Weaver	Senior Product Manager
Reviewer	Sam Kerr	Principal PM
Reviewer	Taylor McCaslin	Group Manager, Product - ModelOps & Anti-Abuse
Reviewer	Viktor Nagy	Senior Product Manager

Meetings

Meetings are recorded and available on YouTube in the Working Group - Product CDF playlist. The playlist is public, although, some meetings maybe private due to the nature of the material covered in the call.

SECTION: company/working-groups/product-development-flow.md

Attributes

Property	Value
Date Created	August 31, 2020
Target End Date	January 30, 2021
Slack	wgproductdevelopmentflow (only accessible by GitLab team members)
Google Doc	Working Group Agenda (only accessible by GitLab team members)
Docs	TBD
Epics/Issues	Main Epic / Issue Board
Label	~wg-product-development-flow
Associated KPIs/OKRs	TBD
GitLab group for working group reviewers	@wg-product-development-flow
GitLab group for working group contributors (MR authors)	@wg-contributors-pd-flow

Problems To Solve

For more details please reference the product development flow survey slides.

Doesn't Encourage Cross-Functional Collaboration

- How can we promote optimal cross-functional involvement in the product development flow (e.g. where would we benefit from having TWs be informed earlier or where can QEs participate earlier)?
- How can we ensure tracks and phases don't silo activities and continually encourage cross-functional contribution everyone can contribute?

Too Many Steps

- How can we reduce the number of "required" steps in the product development flow without losing key steps needed for convergence points (such as workflow states and release notes aggregation)?
- How can we better enable teams to work efficiently by providing the right amount of details in tandem with opportunity for personalization for their specific needs (e.g. an activities/outcomes model rather than just a step by step)?

Too Prescriptive

- How can we reduce verbosity in the product development flow (e.g. provide a better balance of visual snapshots and written descriptions)?
- How do we better focus teams on delivering value by enabling them to strike their own balance between when to build new features and when to improve existing features (e.g. detail out the Improve phase in an alternate style as a test run)?

Business Goals

Iterate on the product development flow to shift away from a "step by step" and more toward being a container/playbook of strategies/tactics to provide teams with:

1. Various activities/outcomes to employ at each phase to identify what to build or improve to solve a problem.
1. Minimal, required processes teams need to follow at each phase to maintain efficient,

transparent and predictable workflow cross-functionally.

Protocols and Processes

Creating Epics & Issues for the working group board

To propose an idea or solution that addresses one of the defined problems - follow the below steps to ensure effective asynchronous collaboration:

1. Create an issue to address your proposal and add it to Product Development Flow as a Framework epic

1. We use this issue board as a single source of truth. Issues should be moved through the board from left to right

1. Assign newly created issues to the Facilitator who is currently @fseifoddini for prioritization

MR review process

- Contributors will generate MRs and tag the Contributor group '@wg-contributors-pd-flow' for feedback first. The Facilitator and Contributors will then review and align if the MR is mergeable (non-controversial) or if it needs more feedback before the merge
- Next, the Facilitator or the MR author will tag in specific people and/or tag in the whole Reviewer group (which includes Executive Sponsors) '@wg-reviewers-pd-flow' for feedback before the merge
 - If unique action is needed from any individual, especially Executive Sponsors, they will be directly pinged. Otherwise, getting pinged as a Reviewer is a standard review/feedback before the merge cycle.
- Our goal is to minimize noise for Reviewers (inclusive of Executive Sponsors) and focus feedback efforts to move forward efficiently. If you want to follow every issue/MR, you're welcome to do so via our WG board

NOTE: Please pay attention to your To-Do list! We'll not hold the train on MRs for more than 24 hours, specially non-controversial ones.

Exit Criteria

Review and Optimize the current Product Development Workflow

- Define the Required and Optional workflow phases.
- Optimize the existing labels currently required in each phase.
- Review the finalized Outcomes/ Activities section in different workflows to incorporate different issue types - feature, bug, experiment, etc.
- Build in early engagement and collaboration with all counterparts in each phase as needed.

Define what success looks like for the PD-FLOW

- Define goals and supporting metrics
- Define qualitative as well as quantitative ways to track, measure and iterate

Dogfooding

- Identify a Section/ group to dogfood the new flow.
- Document and iterate on suggestions and feedback.

Downstream Process Review

- Identify any downstream process changes that might need review.
- Quad Planning process

Matrix and Dashboards

- Document all dashboards that need to be updated.
- Define what matrix should be measured and how.
- Identify any new dashboard needs.

Communication

- Communicate the new product development flow change broadly.
- Conduct an AMA to answer all questions / concerns.

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	Anoop Dawar	VP, Product Management
Executive Sponsor	Christie Lenneville	VP of UX
Facilitator	Farnoosh Seifoddini	Principal PM, Product Operations
Functional Lead	Vincy Wilson (QE)	Manager, Quality Engineering - Growth & Govern
Functional Lead	Jackie Bauer (UX/Product Design)	UX Manager, Enablement & Growth
Functional Lead	Jeff Crow (UX Research)	Senior UX Researcher, Growth
Functional Lead	Michael Karampalas (Product Management)	Principal PM, Growth
Functional Lead	Justin Farris (Product Management)	GPM, Plan
Functional Lead	Craig Norris (Technical Writing)	Technical Writing Manager
Functional Lead	Thomas Woodham (Engineering)	Engineering Manager, Secure
Member	Wayne Haber (Engineering)	Director Engineering, Threat Management
Member	Nadia Udalova (Dev)	Product Design Manager
Member	Keanon O'keefe (Plan)	Senior Product Manager
Member	Tim Hey (Growth)	Principal Product Manager
Member	Eric Schurter (Create)	Senior Product Manager
Member	Mark Wood (Plan)	Senior Product Manager
Member	Sarah Waldner (Monitor)	Senior Product Manager
Member	Fabian Zimmer (Geo)	Senior Product Manager
Member	Daniel Croft (Release, Package)	Senior Manager, Engineering
Member	Jason Yavorska (CI/CD)	Director Product Management
Member	Thiago Figueiró	Engineering Manager, Threat Management
Member	Dan Jensen (Manage)	Engineering Manager, Analytics & Compliance
Member	TBD (Data)	TBD

Working Groups

Working Group	Person	Title
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Contributor	Farnoosh Seifoddini Principal PM, Product Operations
Contributor	Wayne Haber (Engineering) Director Engineering, Threat Management
Contributor	Vincy Wilson (QE) Manager, Quality Engineering - Growth & Govern
Contributor	Jacki Bauer (UX/Product Design) Product Design Manager, Growth & Fulfillment
Contributor	Michael Karampalas (Product Management) Principal PM, Growth
Contributor	Justin Farris (Product Management) GPM, Plan
Contributor	Nadia Udalova (Dev) Product Design Manager
Contributor	Mark Wood (Plan) Senior Product Manager
Contributor	Fabian Zimmer (Geo) Senior Product Manager
Contributor	Daniel Croft (Release, Package) Senior Manager, Engineering
Reviewer	Anoop Dawar VP, Product Management
Reviewer	Christie Lenneville VP of UX
Reviewer	Jeff Crow (UX Research) Senior UX Researcher, Growth
Reviewer	Craig Norris (Technical Writing) Technical Writing Manager
Reviewer	Keanon O'keefe (Plan) Senior Product Manager
Reviewer	Tim Hey (Growth) Principal Product Manager
Reviewer	Eric Schurter (Create) Senior Product Manager
Reviewer	Sarah Waldner (Monitor) Senior Product Manager
Reviewer	Jason Yavorska (CI/CD) Director Product Management
Reviewer	Thiago Figueiró Engineering Manager, Threat Management
Reviewer	Dan Jensen (Manage) Engineering Manager, Analytics & Compliance

Meetings

Meetings are recorded and available on YouTube in the Working Group - Product Development Flow playlist. Due to the subject matter of this working group and the high probability that every synchronous meeting will discuss sensitive customer information, the playlist is private and accessible by GitLab team members only.

SECTION: company/working-groups/purchasing-reliability.md

Attributes

Property	Value
Date Created	2021-12-01
Target End Date	2022-07-29
Slack	wgpurchasingreliability (only accessible from within the company)
Google Doc	Purchasing Reliability Working Group Agenda (only accessible from within the company)
Issue Label	~WorkingGroup::Purchasing

Business Goals

A purchase and ordering system that is well-documented and easily understood by all

stakeholders.

The system will be resilient, reliable, and observable from the ground up.

In addition to addressing our current use-cases, the system will easily accommodate the addition

of new purchasing workflows (like channel partners), metered billing, and project Horse.

Finally, this should all be achieved without sacrifices to the buyer experience. Touchpoints with

the product(s) should be even more seamless than they are today, especially if/as we begin surfacing

up-sell opportunities directly in the product.

Definitions

What is Purchasing?

At GitLab, "fulfillment" generally refers to the Fulfillment Sub-department of Engineering.

To avoid confusion, this Working Group uses "purchasing" as an all-encompassing term to describe

the end-to-end purchasing experience including (but not necessarily limited to) ordering, billing,

licensing, fulfillment, and all of our internal workflows

(like lead generation for Sales, and reporting for Finance).

Be aware if you see the term "fulfillment" used casually in the context of this Working Group, it is likely referring to the entire "purchasing system", not the Fulfillment Sub-department specifically.

Related Projects & Documentation

Slack channels

- sfulfillment
- sfulfillmentengineering
- enterprise-apps
- bt-integrations

Prior Art

- Commercial & Licensing Working Group
 - Handbook Page
 - Agenda Doc
 - Purpose: Define business & customer experience requirements for a system to handle commercial and licensing transactions into the future.
 - Outcomes:
 - Zuora Single Source of Truth Discusssion Paper
 - ???

Infrastructure

Infrastructure team is committed to supporting Fulfillment in the migration of CustomersDot from the legacy Azure VM to GCP by the end of 2022-01. See the discussion in 2021-11-30 Engineering Allocation Review) for details.

This will help bring CustomersDot in line with the rest of our application stack in terms of how it is operated, but the application will still need to be instrumented and alerting configured to address our current lack of observability and insight into this app.

Engineering (Fulfillment)

- CustomersDot
- gitlab-org&3602: GitLab <> Customers, License, Zuora Integration

Enterprise Apps

- Quote to Cash
- Platypus
- gitlab-com/business-technology/enterprise-apps&293: Distributor e-Marketplace

Sales

- Sales Systems (i.e. Salesforce)
- Internal Sales Handbook Definitions
- Booking Metric Technical Docs

Exit Criteria

The charter of this working group is to obtain alignment between stakeholders (i.e. Engineering (Development), Enterprise Apps, Sales, and Finance). This includes addressing some immediate-term technical debt as well as defining a clear vision and team/service boundaries going forward.

1. Infrastructure roadmap for each system defined
1. Data models for each system including sources of truth defined
1. System integrations of Order to Cash systems defined
1. Change management mechanisms across teams defined
1. FY23 team structure and staffing plans for each team defined, with a view into committed roadmap items ~2 quarters out for cross-team coordination.

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	Justin Farris	Sr Director, Product Monetization

Facilitator	Omar Fernandez	Acting Director of Product, Fulfillment
Facilitator	Jerome Z Ng	Senior Engineering Manager, Fulfillment
Member	Steve Loyd	VP Infrastructure
Member	Bryan Wise	VP of IT
Member	Robert Rea	Senior Director, IT Ops
Member	Chase Southard	Engineering Manager, Fulfillment:Utilization
Member	James Lopez	Engineering Manager, Fulfillment:License
Member	Ragnar Hardarson	Engineering Manager, Fulfillment:Purchase
Member	Etienne Baqué	Senior Backend Engineer,
Fulfillment:Utilization		
Member	Tyler Amos	Staff Backend Engineer, Fulfillment:License
Member	Qingyu Zhao	Senior Backend Engineer, Fulfillment:Purchase
Member	Tatyana Golubeva	Principal Product Manager,
Fulfillment:Purchase		
Member	Daniel Parker	Senior Integrations Engineer, Business
Technology		
Member	Mark Quitevis	Senior Business Systems Analyst, Business
Technology		
Member	Courtney Meddaugh	Senior Business Systems Analyst, Business
Technology		
Member	Jack Brennan	Senior Director, Sales Systems
Member	Vincy Wilson	Manager, Quality
Member	Chloe Liu	Senior Software Engineer in Test, Quality
Member	Christopher Nelson	Sr Director, Enterprise Applications

SECTION: company/working-groups/python-stewardship.md

Attributes

Property	Value	
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Date Created	2024-11-01	
Target End Date	2025-03-01	
Epic	https://gitlab.com/groups/gitlab-org/-/epics/15580	
Slack	wgpython-stewardship (only accessible from within the company)	

| Google Doc | Agenda (<https://docs.google.com/document/d/1gs-OrjjyfxQ3BDaKxOXcrMuUl3z1jmsxGmbdmEgIBF8/edit?tab=t.dma9z3zh8fwb>) (only accessible from within the company) |
| Meeting Calendar | Calendar (<https://calendar.google.com/calendar/u/0?cid=Y18xZWE4ZTViZWZiYmUzMDk0MDgzNGJiZWVlMWY1NTFiodVjNWQ0NzQwZDc0MzJhMWQyMDkzOWQ4MzU0YjhkNjU3QGdyb3dsZS5jb20>) (only accessible from within the company) |

Goal

- Consolidate Python development within GitLab, providing confidence to developers on developing high-quality Python codebases.
- Simplify developer decision making by providing opinionated guidelines on how to create, maintain and deploy Python services and libraries in GitLab.
- Define resources (courses, mentorship) for training developers that want or need to contribute to a Python codebase.

Context

While Python has been present at small scale for many years in the company, the evolution of our AI Infrastructure lead to Python becoming a critical part of our product. Yet, we haven't consolidated a development culture around it.

As a result, developers are confused on how to write Python code, how to review Python code, and how to set up new functionality. Ruby developers also feel a lack of support when it comes to understanding and contributing to the codebase. One key difference between our Ruby codebase and our Python codebase is that the Python codebase is more distributed: instead of having a single repository we have multiple each responsible for an area.

Examples of existing Python codebases:

- AI Gateway <https://gitlab.com/gitlab-org/modelops/applied-ml/code-suggestions/ai-assist/>
- Prompt library <https://gitlab.com/gitlab-org/modelops/ai-model-validation-and-research/ai-evaluation/prompt-library>
- Duo Workflow service <https://gitlab.com/gitlab-org/duo-workflow/duo-workflow-service>

Key milestones

- 17.7 - Kick off of Working group
- 17.8 - Contributions to documentation
- 17.9 - Define templates for Python services, libraries, linting and utilities

Exit Criteria

| Criteria | DRI |
| :---- | :---- | :---- |
| A non-Python developer has guidance on how to understand and contribute to Python codebase | @mhamda |
| Guidelines and templates for Python codebase creation (as described in outcomes) have been merged | @tlegitlab |
| Code review and maintainership guidelines for Python codebase have been merged | @brytania |

| Guidelines for deployment of each of the three application types | @eduardobonet |

Outcomes

Documentation and process:

- Training for non-Python developers
- Development guidelines
 - Different types of repositories:
 - Python services (AIGW, workflow)
 - Python utilities (CEF)
 - libraries
 - Creation of codebase
 - Authentication
 - Testing guidelines
- Code review and maintainership guidelines
- Deployment guidelines
 - Python services (AIGW, workflow)
 - Python utilities (CEF)
 - libraries

Development experience:

- Extraction of common setup
 - Linters
 - Frameworks
 - Code Structure
 - CI for deployment

Roles and Responsibilities

| Working Group Role | Person | Title |

|-----|-----|-----|

| Executive Sponsor | Tim Zallmann | VP of Engineering |

| Facilitator | Eduardo Bonet | Staff Fullstack Engineer, Custom models |

| Member | Fred de Gier | Staff Fullstack Engineer, MLOps |

| Member | Alejandro Rodríguez | Senior Backend Engineer, AI Framework |

| Member | Alexander Chueshev | Staff ML Engineer, AI Framework |

| Member | Tan Le | Staff ML Engineer, Model Validation |

| Member | Stephan Rayner | Senior ML Engineer, Model Validation |

| Member | Tetiana Chupryna | Senior Backend Engineer, Duo Chat |

| Member | Mohamed Hamda | Senior Backend Engineer, Custom models |

| Member | Dylan Bernardi | Backend Engineer, Editor Extensions |

| Member | Shola Quadri | Associate Backend Engineer, Code Creation |

| Member | Vitali Tatarintev | Senior Backend Engineer, Code Creation |

| Member | Miko·aj Wawrzyniak | Staff Backend Engineer, Duo Workflow (Borrow) |

| Member | Ryan Egesdahl | Senior Distribution Engineer, Distribution Build |

Sync Recordings

- <https://www.youtube.com/playlist?list=PL05JrBw4t0KpgQYHyThWFSHkRQWoDgcWT>

Non-goals

- Evolution of AI Gateway, although related to this WG proposal, is out of scope.
- Data Science use cases (such as code to develop a new ML model) differs substantially to system development, and merits their own set of guidelines.

SECTION: company/working-groups/rate-limit-architecture.md

Attributes

Property	Value
Date Created	2022-04-27
Target End Date	2022-12-09
Slack	wgratelimitarchitecture (only accessible from within the company)
Google Doc	Agenda (https://docs.google.com/document/d/1nv8hDLqZS16yTa4M3FtdMQNOFM4dt4OXBGCPvp-FHbM/edit) (only accessible from within the company)

Summary

The Rate Limit Architecture WG was active from April - December 2022. In FY23-Q1 we regularly experienced site incidents related to a lack of rate limits on various features. We initiated an Engineering Allocation for Rate Limiting Across DevOps Platform which performed a Rate Limiting audit for each DevOps Stage, and updated development docs and guidelines to provide clearer rate limit related guidance, and closed 70+ rate limiting issues. This resulted in increased reliability and a major reduction in the number of rate limit related site incidents.

Following the Engineering Allocation we initiated this Working Group to establish a forward looking vision for Rate Limiting. The group developed a Rate Limiting Architecture Blueprint which captures this vision. The group also collaborated to identify which GitLab stage group would own this functionality long term. This was documented by adding the Rate Limiting Framework to the Application Performance group. The blueprint will continue to be reviewed regularly as part of the Architecture Evolution Workflow process.

As the Working Group has accomplished its exit criteria we made the decision to disband the group.

Context

During FY23-Q1 we prioritized an engineering allocation for Rate Limiting Across DevOps Platform focused on identifying and addressing denial of service (DoS) vectors via rate and application limits. So far that effort has been successful in addressing 70+ of DoS related issues.

In addition to addressing DoS vectors to ensure high availability we have goals around:

- consistent monitoring and alerting on limits
- consistent, comprehensive documentation of limits
- synchronization of GitLab.com limit settings with GitLab default limits

Currently there are multiple approaches to rate limiting documented in GitLab application limits development documentation.

Exit Criteria

This Working Group has the following goals:

1. Develop a recommendation for the preferred way to implement application rate limits within GitLab
1. Document this recommendation for GitLab developers in an Architecture Evolution Blueprint.
1. Document iterative project plan for short term, medium term, and long term rate limiting improvements
1. Determine organizational approach to funding effort on this work (e.g. new team, borrow, SIG, fan out, etc.)

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	Christopher Lefelhocz	VP of Development
Facilitator	Sam Goldstein	Director Engineering, Ops
Member	Grzegorz Bizon	Principal Engineer, Verify
Member	Fabio Pitino	Staff Backend Engineer, Verify
Member	Marshall Cottrell	Strategy and Operations (Technical)
Member	Hayley Swimelar	Senior Backend Engineer, Container Registry
Member	Quang-Minh Nguyen	Senior Backend Engineer, Scalability
Member	Liam McAndrew	Engineering Manager, Scalability
Member	Markus Koller	Senior Backend Engineer, Integrations
Member	Grant Hickman	Sr Product Manager, Integrations
Architecture Evolution Coach	Andrew Newdigate	Distinguished Engineer, Infrastructure

 ## SECTION: company/working-groups/recruiting-ssot.md

Attributes

Property	Value
Date Created	December 30, 2019
End Date	TBD
Slack	wgrecruiting-ssot (only accessible from within the company)
Google Doc	Talent Acquisition SSOT WG Agenda (only accessible from within the company)

Business Goal

To create a Talent Acquisition SSOT and a reconciliation process to use with the Talent Acquisition SSOT. The Talent Acquisition SSOT will be used by Finance for forecasting and Talent Acquisition for hiring. It will include all forecasted hiring for the fiscal year that is agreed upon by Finance and the GitLab E-Group.

Exit Criteria

- Create a Talent Acquisition SSOT document that has all the basic information needed for Talent Acquisition to post reqs for the fiscal year and be able to contact hiring managers for additional detail
- Be able to reconcile GitLab's talent acquisition platform with the Talent Acquisition SSOT on a monthly basis
- Be able to reconcile recent hires with Talent Acquisition SSOT on a monthly basis
- Create processes for when changes are required from the Talent Acquisition SSOT

Roles and Responsibilities

Working Group Role	Person	Title
Facilitator	Christine Machado	Finance Business Partner
Functional Lead	Dave Gilbert	VP, Talent Acquisition
Member	Erich Wegscheider	Talent Acquisition Program Analyst
Member	Chase Wright	Manager, Financial Planning & Analysis
Member	Kelly Murdock	Manager, Talent Acquisition
Member	April Hoffbauer	Manager, Talent Acquisition Operations
Member	Kathleen Tam	Manager, Data
Executive Stakeholder	Craig Mestel	VP, Finance

SECTION: company/working-groups/runtime-update-process.md

Attributes

Property	Value
Date Created	2022-11-29
Target End Date	TBD
Slack	wg-runtime-update-process
Google Doc	Working Group Agenda
Task Board	Issue board
Epic	Link

Business Goal

We need to streamline the language framework (version) update process. Language framework updates are driven by multiple teams, cumbersome, and the processes are inconsistent between the multiple frameworks we use today. In the case of Go, a half dozen or more teams are involved. The current upgrade process is difficult to execute at scale and requires manual intervention with a high degree of complexity. Goals include, but are not limited to:

- Create a streamlined Golang runtime (version) update process or possibly a language framework agnostic process
- Reduce update process complexity
- Create documentation around the update process to improve efficiency
- Increase developer and management efficiency

Tracking Progress

Progress will be tracked on the Working Group issue board using the following labels:

- ~"workflow::In dev"
 - The issue is currently in progress and actively being worked on
- ~"workflow::In review"
 - The issue is currently being reviewed by broader Engineering Team
- ~"workflow::blocked"
 - The issue is blocked by another issue. Please refer to the blocking issue.
- ~"workflow::production"
 - The issue has been completed and should be closed.

Exit Criteria (0% completed)

Criteria	Start Date	Target Completion Date	Completed Date	DRI
Build and document a repeatable upgrade process that's applicable to our major platforms such as Go, Ruby, Rails, and Vue.	2023-03-30	2023-08-31		
Upgrade to Go 1.20	2023-03-30	2023-07-30		@rmarshall
Evaluate a second framework for an upgrade such as Ruby, Python, or Node	2023-06-01	2023-06-30		
Upgrade a second framework to validate our process	2023-07-01	2023-09-30		

Roles and Responsibilities

Working Group Role	Person	Title
Executive Sponsor	Christopher Lefelhocz	VP of Development
Facilitator	Matt Nohr	Engineering Manager, Create
Functional Lead	Robert Marshall	Senior Backend
Member	Chun Du	Director of Engineering, Enablement
Member	Thomas Woodham	Senior Engineering Manager, Secure
Member	Ross Fuhrman	Senior Backend
Member	Patrick Bajao	Senior Backend Engineer, Create

SECTION: company/working-groups/secure-offline-environment.md

Attributes

Property	Value
Date Created	March 17, 2020
Target End Date	April 22, 2020
Slack	wgsecure-offline-environment (only accessible from within the company)
Google Doc	Secure Offline Deployment Working Group Agenda (only accessible from within the company)
Tracking Epic	gitlab-org Epic

Business Goal

Allow mirroring or downloading of all the needed information for an offline use. Data will be replicated and kept in sync on a separate server, internal to the same network where the runners are, and will provide all the information needed. Provide documentation & instructions describing how to configure an offline environment instance to run scanners successfully.

Exit Criteria

- Score a 5.0 on the demo scorecard with released code by April 22, 2020

Roles and Responsibilities

Working Group Role	Person	Title
Facilitator	Todd Stadelhofer	Director of Engineering, Secure
Quality Lead	Mek Stittri	Director of Quality
CSM Lead	Kevin Chasse	Federal Customer Success Manager
PM Lead	Nicole Schwartz	Product Manager, Secure, Composition Analysis
Member	David DeSanto	Director, Product, Sec
Member	Olivier Gonzalez	Engineering Manager, Secure, Composition Analysis
Member	Seth Berger	Engineering Manager, Secure, Dynamic Analysis
Member	Thomas Woodham	Engineering Manager, Secure, Static Analysis
Executive Sponsor	Eric Johnson	Chief Technology Officer

SECTION: company/working-groups/self-managed-scalability.md

Attributes

Property	Value
Date Created	May 21, 2019

Target End Date	June 23, 2020
Slack	wgsml-scalability (only accessible from within the company)
Google Doc	Self-managed Scalability Working Group Agenda (only accessible from within the company)
Issue Board	gitlab-org boards 1131633

Business Goal

Ensure all new customers are set up in a standardized environment that will scales with their needs. Migrate existing customers to an appropriate reference environment.

Exit Criteria (100%)

- Complete GitLab self-managed performance metrics => 100%
- Monitoring enabled by default for all large customers (Prometheus, Grafana and all exporters) => 100%
- Inventory of self-managed customers with scores of their environment => 100% Scoring system has been defined.
- Migrate one customer to our reference architecture => 100%
- List of additional needed reference architectures => 100% We have identified 10K, 25K and 50K and the next immediate need.
- 10,000 user reference architecture => 100%
- 25,000 user reference architecture => 100%
- 50,000 user reference architecture => 100%
- First iteration of a mixed Kubernetes Omnibus architecture => 100%

Business outcome

- Since we have completed the performance testing, Our reference architecture has been the default starting point for self-managed customers looking to set up HA. This is a significant improvement in efficiency by reducing the time SAs spend helping customers determine the right sizing/setup.
- The performance testing kit has been utilized by prospects who need to prove performance at scale as part of their pre-purchase due diligence. Previously, the customer had to do this on their own, which was time consuming, complicated, and risked the deal moving forward.

Roles and Responsibilities

Working Group Role	Person	Title
Facilitator	Mek Stittri	Director of Quality
Support Lead	Aric Buerer	Support Engineer
CS Lead	Brian Wald	Solutions Architects Manager
Quality Lead	Grant Young	Sr. Software Engineer in Test
Development Lead	Matt Nohr	Engineering Manager, Monitor
PM Lead	Dov Herskovitch	Senior Product Manager, Monitor:APM
Tech Writing Lead	Achilleas (Axil) Pipinellis	Technical Writer
Member	Tom Cooney	Director of Support
Member	Francis Potter	Solutions Architect

Member	John Woods	CSM Manager	
Member	Tanya Pazitny	Quality Engineering Manager	
Member	Andrew Newdigate	Distinguished SRE	
Member	Nailia Iskhakova	Software Engineer in Test	
Member	John Skarbek	Senior Site Reliability Engineer, Delivery	
Member	Jason Plum	Senior Distribution Engineer	
Member	Vincy Wilson	Quality Engineering Manager	
Member	Catalin Irimie	Support Engineer	
Executive Sponsor	Eric Johnson	VP of Engineering	

SECTION: company/working-groups/sharding.md

Attributes

Property	Value	
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Date Created	February 11, 2020	
End Date	June 22, 2020	
Slack	wgdatabase-sharding (only accessible from within the company)	
Google Doc	Sharding Working Group Agenda (only accessible from within the company)	
Recordings	Sharding Working Group Playlist	

Outcome - Closed

We have decided to close this Sharding focused working group and will open a Scaling Working Group with a different focus. The initial focus of this Sharding working group was to increase the scalability of our database with a long-term goal of 100x scalability. At the onset of this group, it was theorized that we would hit a database scalability wall within 6-12 months. Subsequent analysis and incremental scalability efforts have indicated that we have significantly more scaling headroom. Based on the analysis we have a high degree of confidence that the current architecture is in good shape to handle our needs for the next 12 months: Database Capacity and Saturation Analysis (Iteration 1) This analysis will continue on a monthly basis. We have also identified areas of incremental database scalability that has been prioritized by the database team: Reduce total size and growth of GitLab.com's PostgreSQL database. Between the ongoing analysis and incremental database improvements we have greatly reduced the urgency of database scalability.

Additionally, we have come to the consensus that sharding is not the desired approach for our long term scalability needs. This decision was informed through investigation, proofs of concept, research, interviews and various implementation proposals. Here's a brief list of items that helped to inform our decision to close this working group:

- Introducing PostgreSQL table partitioning to GitLab's database
- Table partitioning: Issue group search as an example
- Sharding with CitusDB
- Postgres: Sharding with foreign data wrappers and partitioning
- Sharding GitLab by root namespace
- Database Sharding Proposal: Sharding by Namespace

- An Alternate Database Sharding Proposal: Tenant Sharding

The core members of this working group will continue on with the Scaling Working Group to determine our long term scaling strategy and implementation. The rest of this working group page will remain for reference purposes.

Business Goals

A scalability approach that will give us 100x headroom over what we have now o